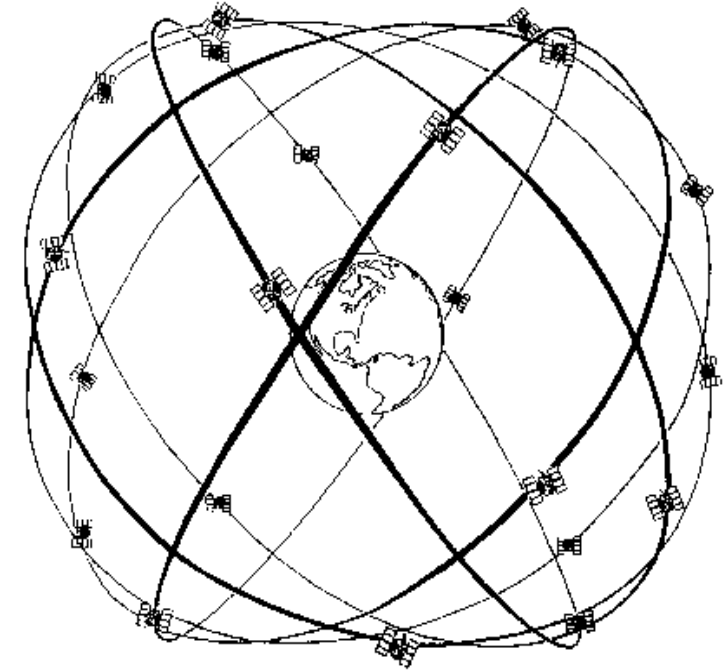
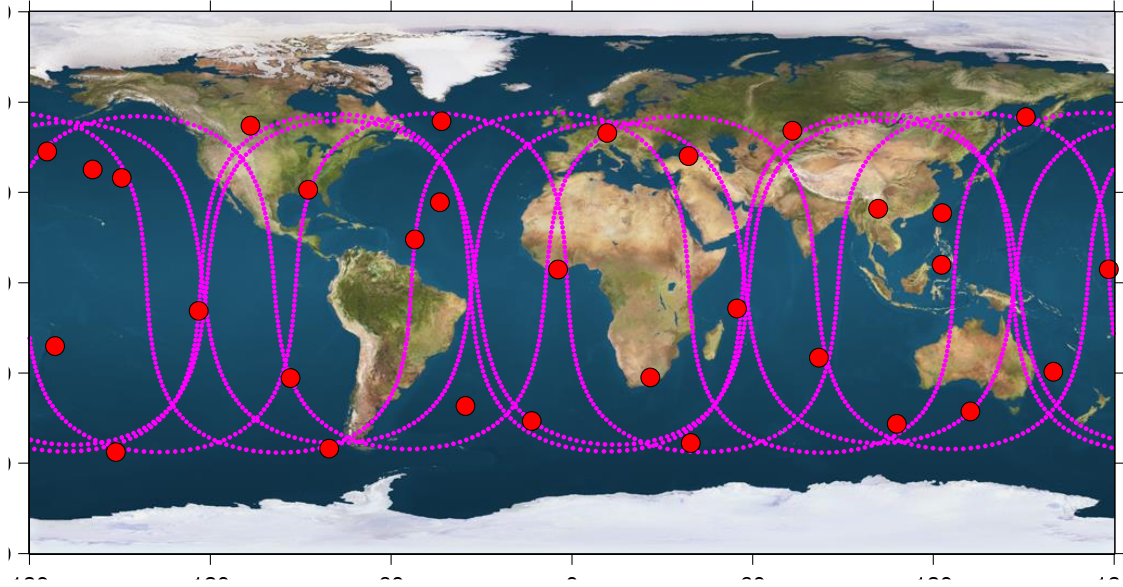


ASEN 5090 Lecture 5

ORBITS, GNSS CONSTELLATIONS, & VISIBILITY



Penny Axelrad

Penina.Axelrad@Colorado.edu

(303) 492-6872



Lectures 6 & 7 will be recorded
Modified Office Hours will be scheduled on ZOOM

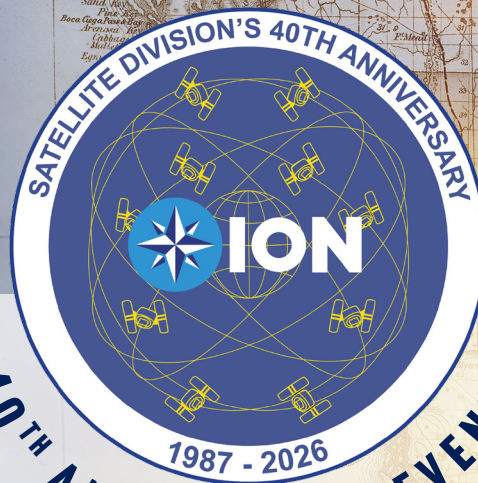


ION GNSS+ 2026

+ SEPTEMBER 14-18, 2026

HYATT REGENCY GRAND CYPRESS

ORLANDO, FLORIDA



40TH ANNIVERSARY EVENTS

EXPANDED MASTERS SHORT COURSE PROGRAM

40TH ANNIVERSARY STUDENT SESSION

TRIVIA EVENT!



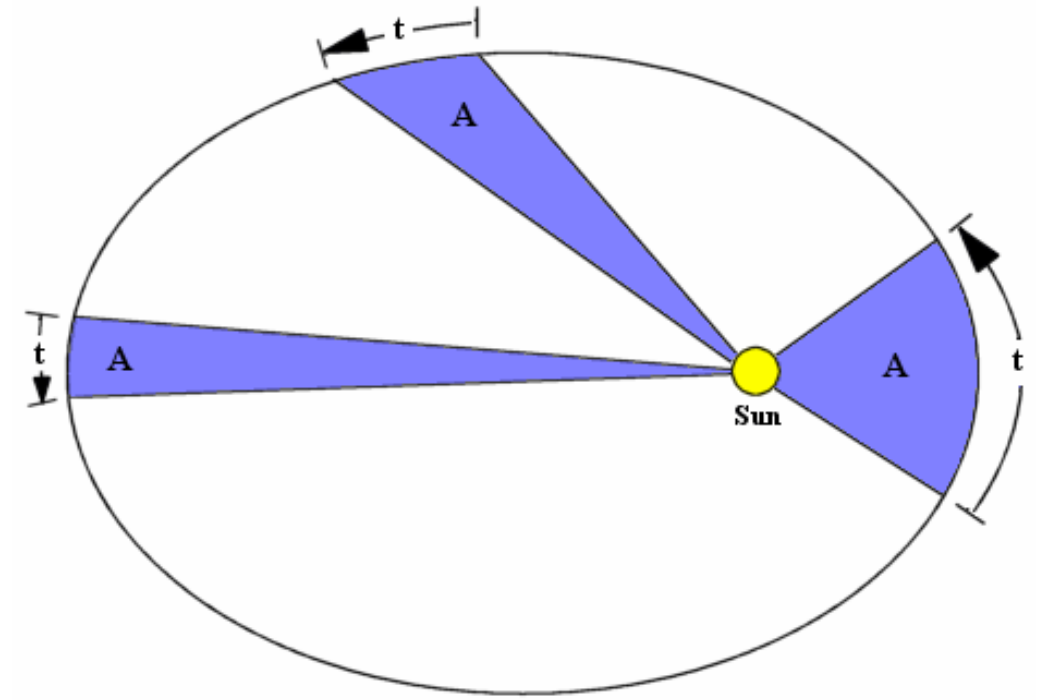
Outline

- Kepler's Laws: 2-body motion
- Orbital Elements
- GNSS Constellations
- Orbital Perturbations
- GPS Ephemerides – Almanac, Broadcast Ephemeris, sp3 files
- Visibility

KEPLER'S LAWS

Describe motion of an object under a central force

1. Planets take the path of an ellipse, with the Sun at one focus.
2. The line joining the planet to the sun sweeps out equal areas in equal times.
3. The square of the orbital period is proportional to the cube of its mean distance to the sun.

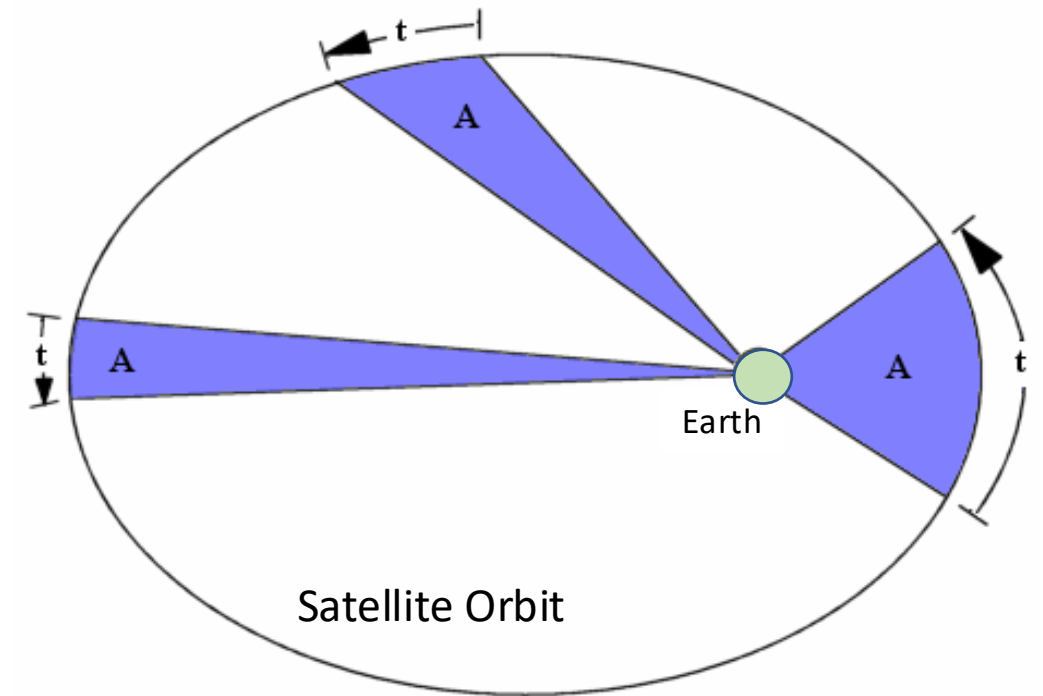


Planet's Orbit

Two-body motion of satellite orbiting a planet

$$\frac{d^2 \vec{r}}{dt^2} = -\frac{\mu}{r^3} \vec{r}$$

This version of the equation of motion is approximate, assuming planet is a point mass and mass of satellite is negligible compared to the planet.

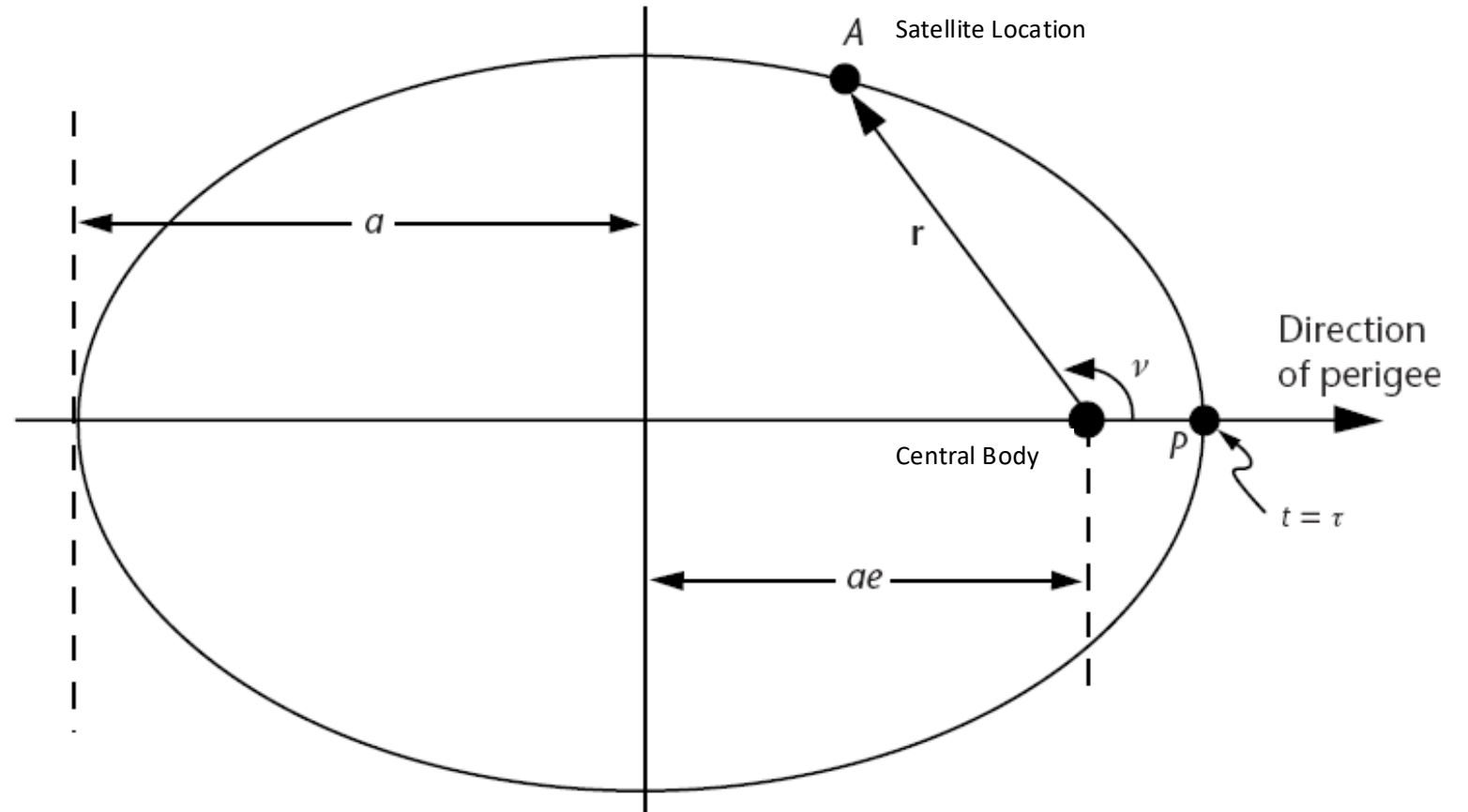


Keplerian Elements

- **Size & shape of the orbit**
- Orientation of the orbit plane
- Position within the orbit

a : semi-major axis

e : eccentricity



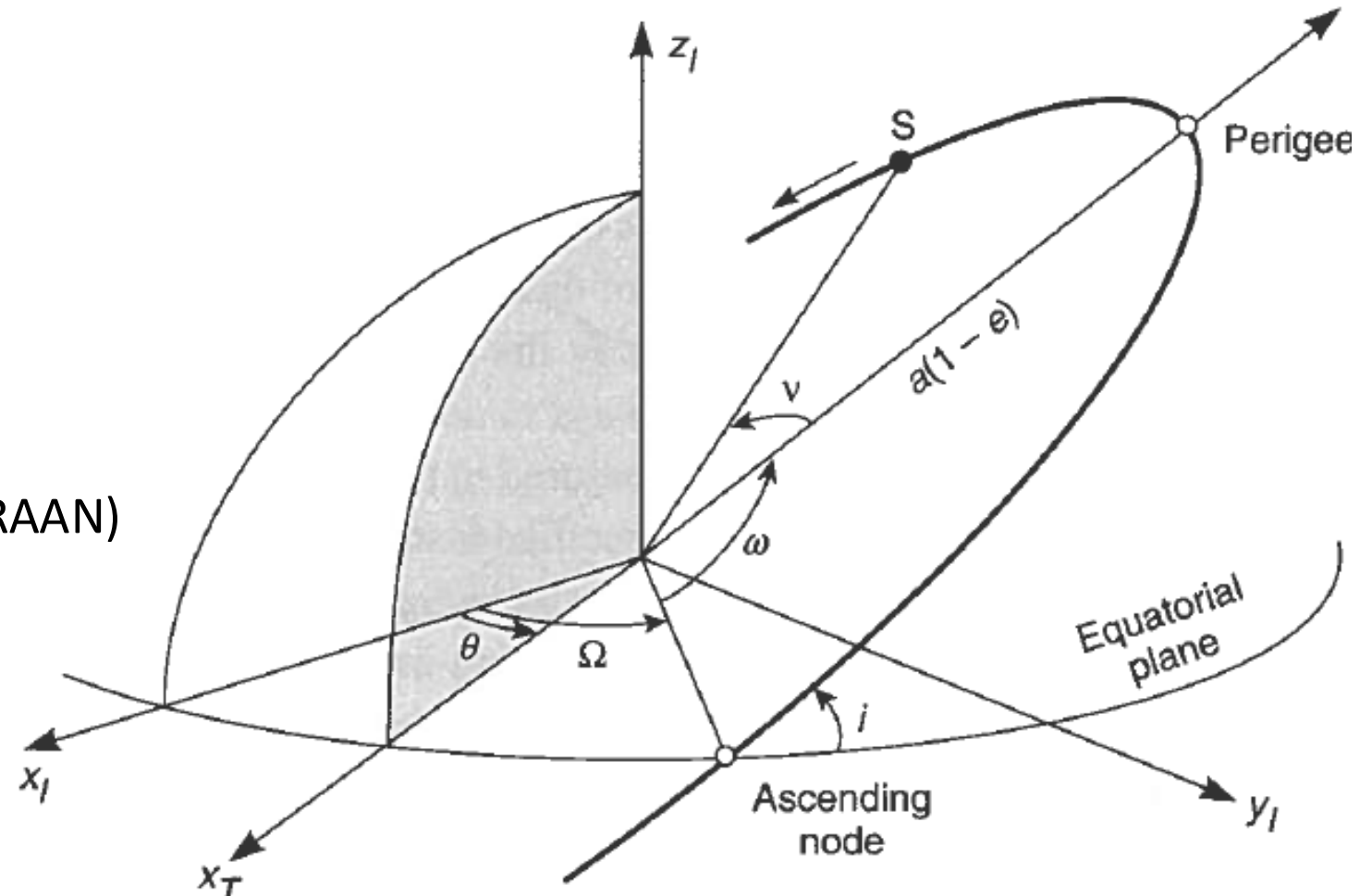
Keplerian Elements

- Size & shape of the orbit
- **Orientation of the orbit plane**
- Position within the orbit

i : orbit inclination angle

Ω : right ascension of ascending node (RAAN)

ω : argument of perigee



Keplerian Elements

- Size & shape of the orbit
- Orientation of the orbit plane
- **Position within the orbit**

(One of the following)

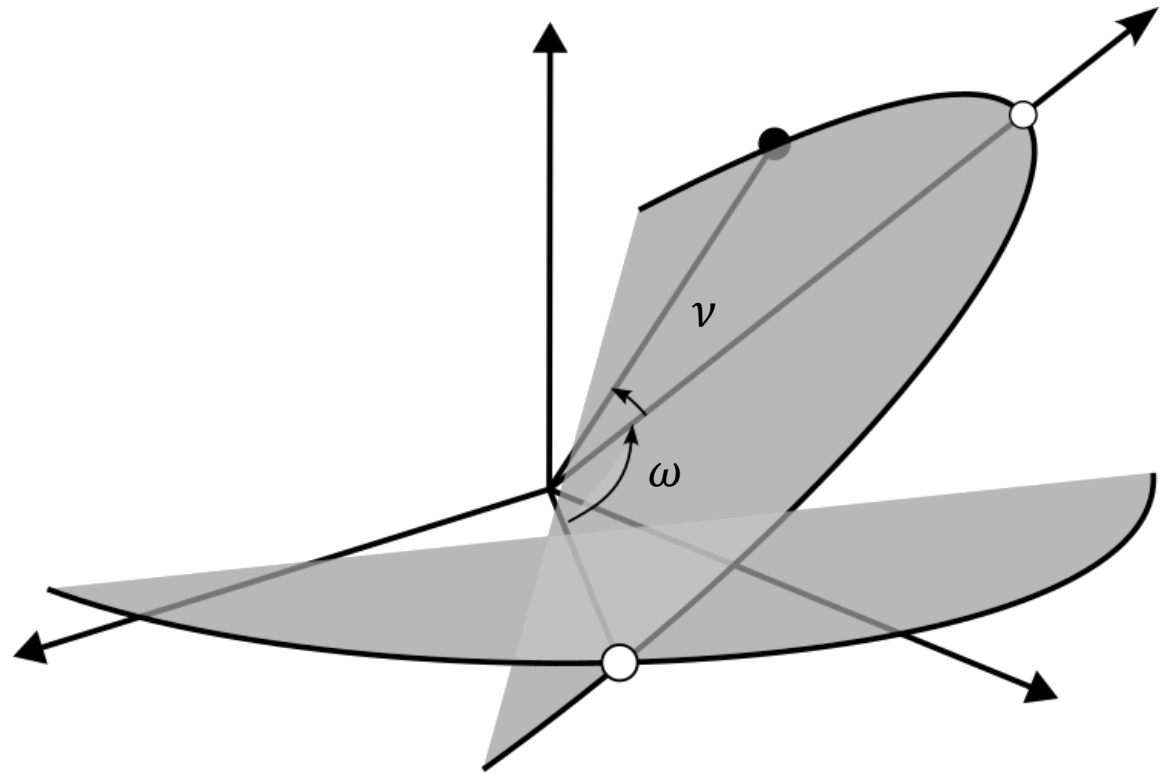
ν : true anomaly

u : argument of latitude = $\omega + \nu$

M : mean anomaly (see next slide)

E : eccentric anomaly (see next slide)

Note: “argument” = “anomaly” = angle



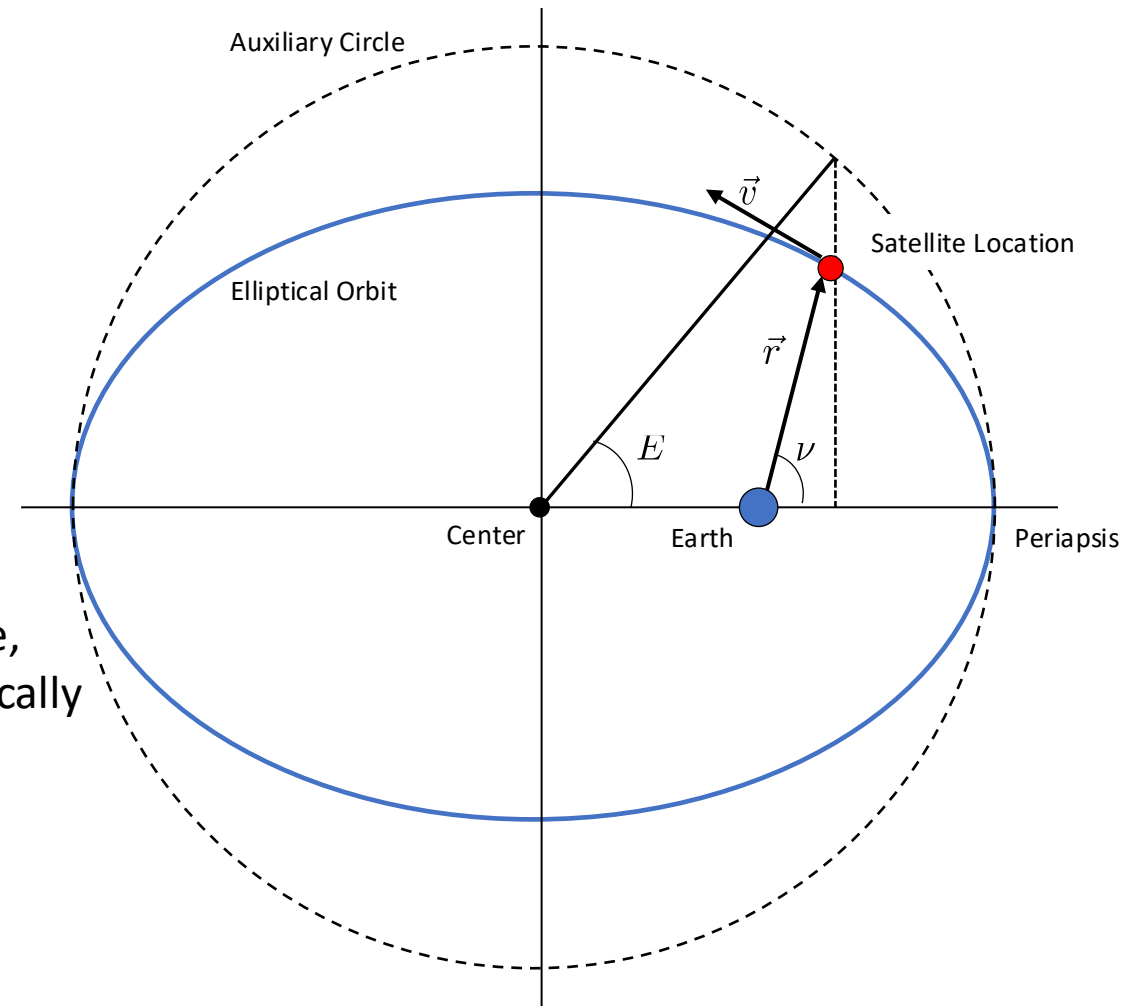
Mean / Eccentric Anomaly

$$P = 2\pi \sqrt{\frac{a^3}{\mu}} \quad \text{orbit period}$$

$$n = \frac{2\pi}{P} = \sqrt{\frac{\mu}{a^3}} \quad \text{mean motion}$$

$$M = M_0 + n(t - t_0) \quad \text{mean anomaly}$$

Not a real angle,
just mathematically
convenient



Kepler's Equation

$$M = E - e \sin E$$

KEPLERIAN ORBITAL ELEMENTS

Size and shape of the orbit:

a : semimajor axis of the ellipse

e : eccentricity of the ellipse $b = a\sqrt{1 - e^2}$

Orientation of the orbit:

Ω : right ascension of the ascending node

i : inclination

ω : argument of perigee

Position within the orbit (pick one):

ν : true anomaly

u : argument of latitude ($\omega + \nu$)

E : eccentric anomaly

M : mean anomaly



POSITION WITHIN ORBIT PLANE

- Given M_0 at a reference time, compute M for time of interest.
- Solve Kepler's equation for E iteratively

- Then compute the true anomaly

$$\sin \nu = \frac{\sqrt{1 - e^2} \sin E}{1 - e \cos E}$$
$$\cos \nu = \frac{\cos E - e}{1 - e \cos E}$$
$$\nu = \tan^{-1} \frac{\sqrt{1 - e^2} \sin E}{\cos E - e}$$

- And the distance from the focus

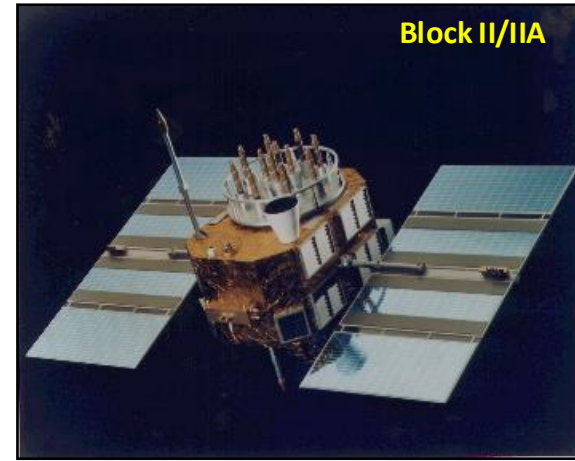
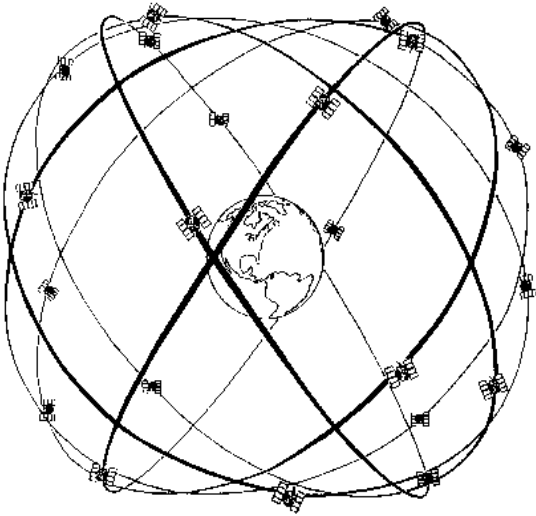
$$r = a (1 - e \cos E) = \frac{a (1 - e^2)}{1 + e \cos \nu}$$

Also useful, is the vis-viva equation for velocity magnitude $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$

GPS CONSTELLATION



GPS Space Segment



- 24-satellite (**nominal**) constellation – 31 active currently
- Six orbital planes inclined at 55 deg, four satellites per plane
- **Not geosynchronous.**
- Semi-synchronous, **nearly** circular orbits ($a=26,560$ km)
- Redundant cesium and/or rubidium atomic clocks onboard each satellite
- Antenna array pointed at the earth

NOMINAL GPS 24-SATELLITE CONSTELLATION

Semimajor axis = 26559.8 km
inclination = 55 deg, eccentricity = 0

Epoch: 23:59:43 UTC, 31 December 2016
(Week 1930 / Time of Week 0000 GPS time)

Greenwich Hour Angle: 100.765°

Referenced to FK5/J2000.00 Coordinates

Orbital Slot IDs are Arbitrarily Numbered

* Orbital Slots Marked by an Asterisk are Expandable

2020 SPS Performance Standard FINAL

Table 3.2-1. Baseline 24-Slot Constellation Slot Assignments as of the Defined Epoch

Slot	RAAN	Argument of Latitude	GEC (GLAN)
A1	288.85°	239.54°	127.85°
A2*	288.85°	133.20°	74.68°
A3	288.85°	343.09°	179.63°
A4	288.85°	13.22°	14.69°
B1*	348.85°	52.37°	94.27°
B2	348.85°	144.75°	140.46°
B3	348.85°	281.39°	28.78°
B4	348.85°	175.79°	155.98°
C1	48.85°	83.29°	169.73°
C2	48.85°	343.21°	119.69°
C3	48.85°	311.08°	103.62°
C4*	48.85°	212.97°	54.57°
D1	108.85°	106.64°	61.40°
D2*	108.85°	236.86°	126.51°
D3	108.85°	6.57°	11.37°
D4	108.85°	138.77°	77.47°
E1	168.85°	168.46°	152.31°
E2	168.85°	274.01°	25.09°
E3*	168.85°	37.48°	86.82°
E4	168.85°	305.10°	40.63°
F1	228.85°	210.30°	53.23°
F2*	228.85°	316.64°	106.40°
F3	228.85°	76.62°	166.39°
F4	228.85°	106.76°	1.46°

Expandable 24-Slot Constellation Slot Assignments

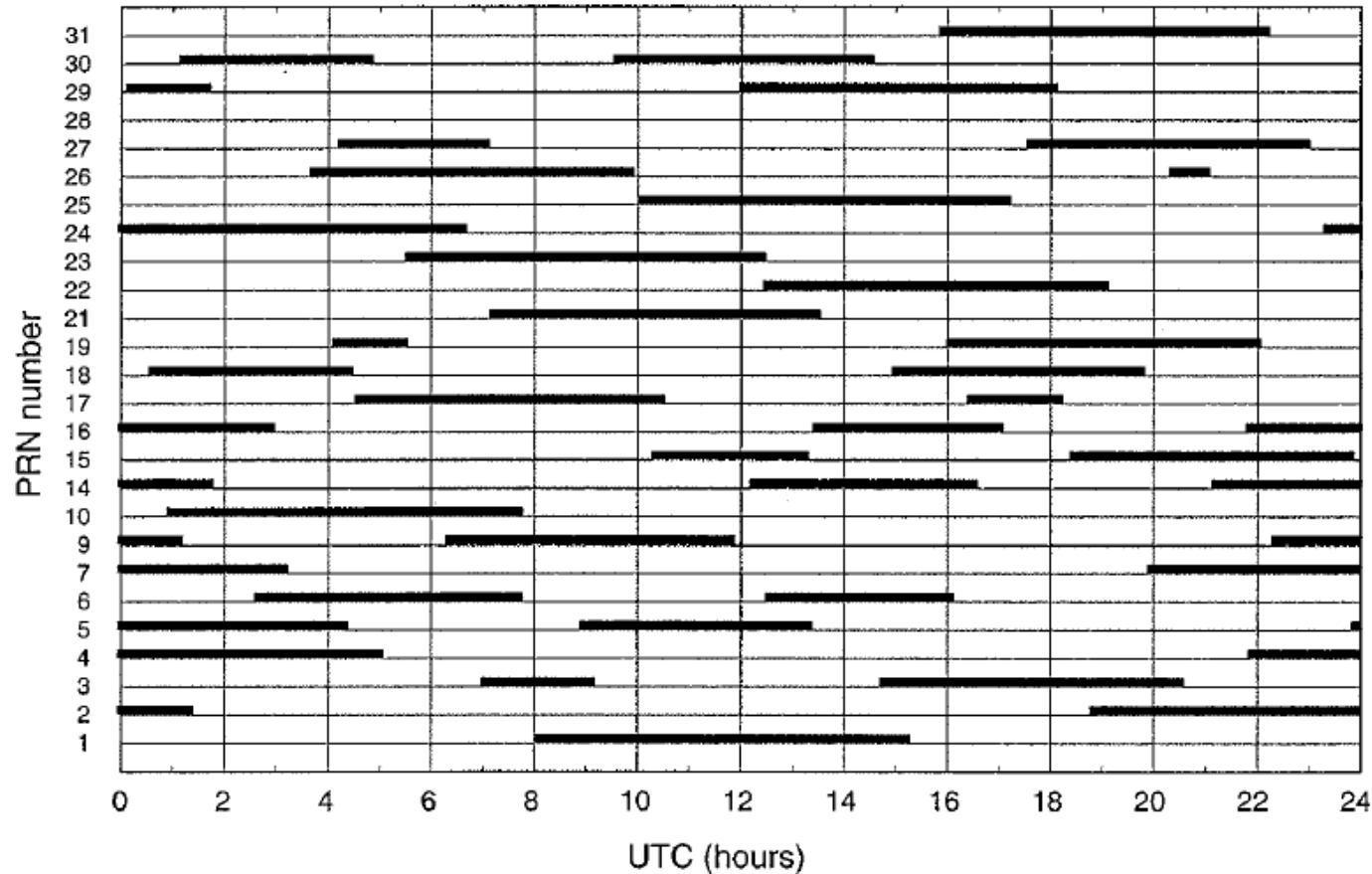
Table 3.2-2. Expandable 24-Slot Constellation Slot Assignments as of the Defined Epoch

Expandable Slot		RAAN	Argument of Latitude	GEC (GLAN)
B1 Expands To:	B1F	348.85°	66.33°	101.25°
	B1A	348.85°	37.77°	86.97°
D2 Expands To:	D2F	108.85°	254.09°	135.13°
	D2A	108.85°	229.39°	122.78°
F2 Expands To:	F2F	228.85°	331.87°	114.02°
	F2A	228.85°	305.43°	100.80°
A2 Expands To:	A2F	288.85°	144.40°	80.28°
	A2A	288.85°	117.96°	67.06°
C4 Expands To:	C4F	48.85°	220.43°	58.30°
	C4A	48.85°	195.73°	45.95°
E3 Expands To:	E3F	168.85°	52.08°	94.18°
	E3A	168.85°	23.52°	79.84°

Groundtrack Equatorial Crossing (GEC) a.k.a.
Geographic Longitude of the Ascending Node [GLAN]

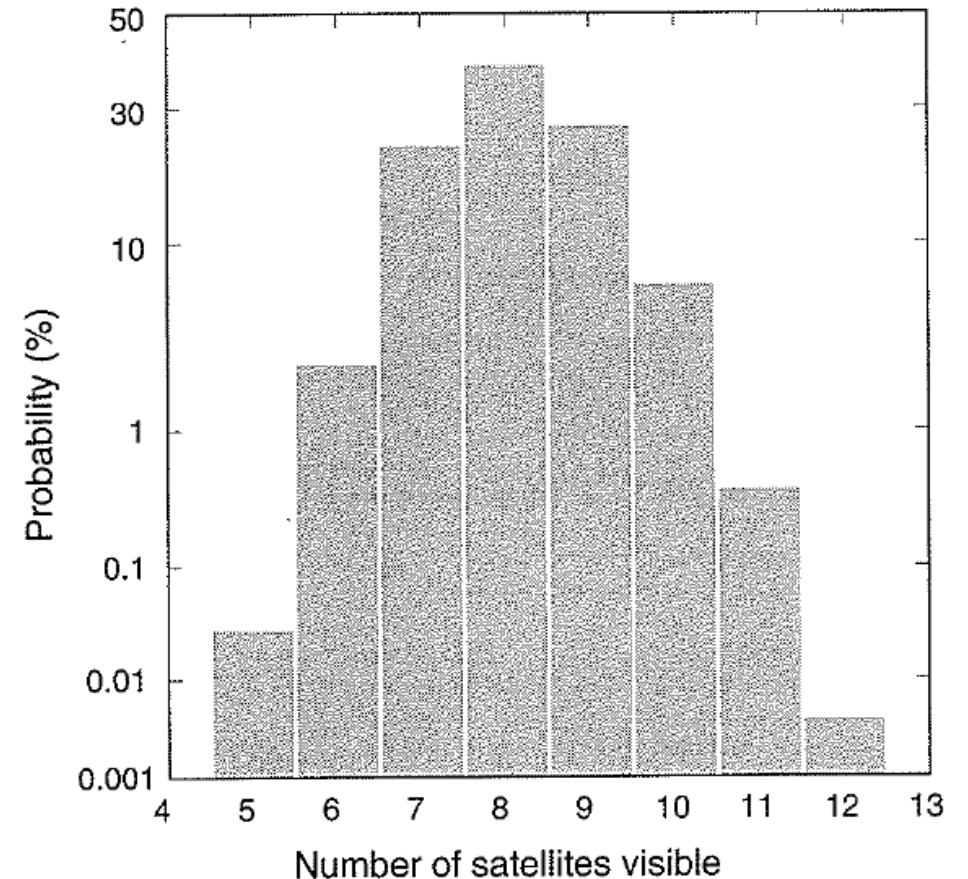
<https://www.navcen.uscg.gov/sites/default/files/pdf/gps/geninfo/2020SPSPPerformanceStandardFINAL.pdf>

Example Satellite Visibility & Statistics



Misra & Enge, P131, Fig. 4.15

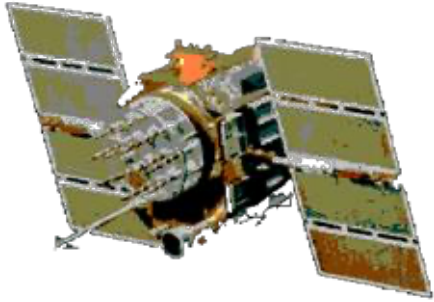
Nominal 24 satellite Constellation,
5-deg elevation mask, Worldwide



Misra & Enge, P132, Fig. 4.17

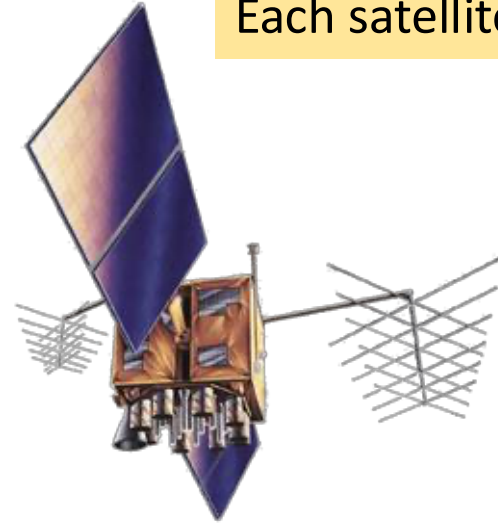
GPS Satellite Blocks & Status (July 2026)

31 active, 1 commissioning, 1 residual



Block II/IIA
Boeing (Rockwell)
Launched 1989 – 1997

Block IIF
Boeing
Launched 2010-2016
11 Operational



Transmit antenna array points toward the Earth
Each satellite carries 3-4 atomic clocks

Block IIR/IIR-M
Lockheed Martin
Launched 1997 – 2009
11 Operational, 1 possibly in residual status

GPS III
Lockheed Martin
Launched 2018 -
9 Operational, 1 commissioning



Check out their nicknames!
SVN79 is *Amelia Earhart*
SVN81 is *Katherine Johnson*

GPS Jargon

- **PRN (pseudorandom noise code)** tells you which code is being generated by a certain satellite.
The codes get reused because for legacy system there are only 32 available for assignment to satellites
- **SVN (satellite vehicle number)** tells you exactly which spacecraft it is.
This number is not reused as it relates to the design specifications of the satellite.
- Constellation status is maintained by the Coast Guard, (NANU)
<https://www.navcen.uscg.gov/?Do=constellationStatus>
- For example, PRN 10 was assigned to
 - SVN40 from 1996-07-16 00:00:00 to 2015-08-03 23:59:59
 - SVN36 from 2015-09-16 00:00:00 to 2015-10-26 23:59:59
 - SVN73 from 2015-10-31 00:00:00 – present
- Mapping of PRNs to SVNs is available here: <https://celestrak.org/GPS/GPSData.txt>

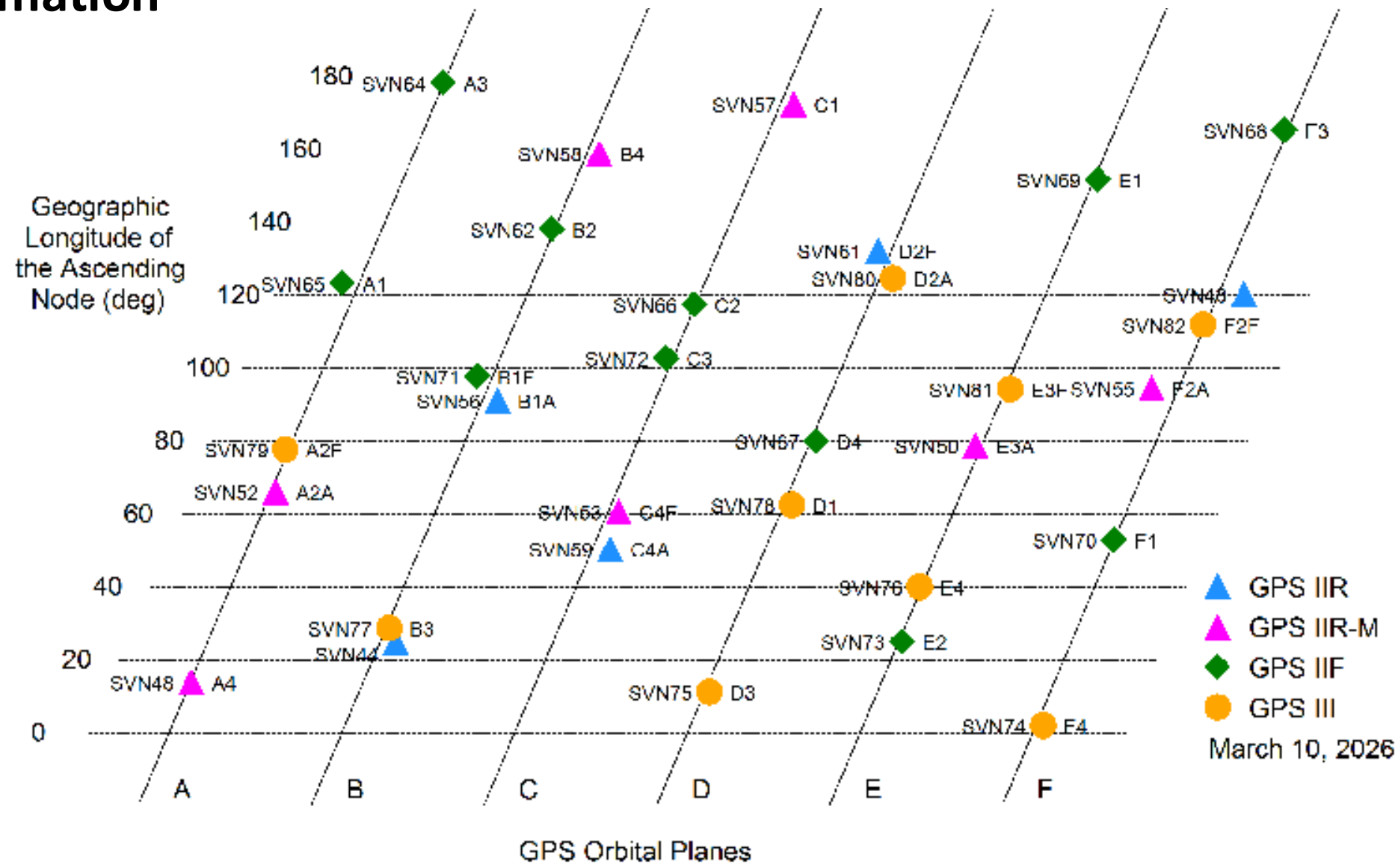
Constellation Status

(navcen.uscg.gov)

GPS CONSTELLATION STATUS FOR Sun Aug 23 2026								
Plane	Slot	SVN	PRN ▲	Block-Type	Clock	Outage Start	NANU Type	NANU Subject
D	2	80	1	III	RB			
D	1	61	2	IIR	RB			
E	1	69	3	IIF	RB			
F	4	74	4	III	RB			
E	3	50	5	IIR-M	RB			
D	4	67	6	IIF	RB			
A	4	48	7	IIR-M	RB			
C	3	72	8	IIF	RB			
F	3	68	9	IIF	RB			
E	2	73	10	IIF	RB			
D	5	78	11	III	RB			
B	4	58	12	IIR-M	RB			
A	5	43	13	IIR	RB	24 APR 202624 APR 2026	DECOM	SVN43 (PRN13) DECOMMISSIONING JDAY 114/2200

Constellation Information

(navcen.uscg.gov)



GNSS Satellite Perturbing Accelerations and Effect on Orbit

From Springer Handbook, Ch3 Hugentobler & Montenbruck

Table 3.2 Average acceleration and orbit error after two revolutions for different orbit types and perturbations

	GPS			Galileo			IGSO		
Perturbation	Average acceleration	Orbit error after 2 rev.		Average acceleration	Orbit error after 2 rev.		Average acceleration	Orbit error after 2 rev.	
		Initial conditions			Initial conditions			Initial conditions	
		Fixed	Adjusted		Fixed	Adjusted		Fixed	Adjusted
	(m/s ²)	(m)	(m)	(m/s ²)	(m)	(m)	(m/s ²)	(m)	(m)
Earth oblateness	5.7 · 10 ⁻⁵	23 000	3000	3.8 · 10 ⁻⁵	22 000	2700	9.1 · 10 ⁻⁶	16 000	1900
Direct tides Moon	3.0 · 10 ⁻⁶	1900	170	3.3 · 10 ⁻⁶	2700	270	4.7 · 10 ⁻⁶	12 000	1100
Direct tides Sun	1.6 · 10 ⁻⁶	930	90	1.7 · 10 ⁻⁶	1700	110	2.5 · 10 ⁻⁶	6900	480
Higher potential terms	3.7 · 10 ⁻⁷	360	32	2.4 · 10 ⁻⁷	340	30	5.6 · 10 ⁻⁸	1100	85
Direct solar rad.press.	1.0 · 10 ⁻⁷	220	32	1.0 · 10 ⁻⁷	290	44	1.0 · 10 ⁻⁷	860	130
Earth albedo	9.8 · 10 ⁻¹⁰	1.1	0.050	1.4 · 10 ⁻⁹	2.2	0.11	7.0 · 10 ⁻¹⁰	3.1	0.15
Solid Earth tides	1.1 · 10 ⁻⁹	0.70	0.044	7.4 · 10 ⁻¹⁰	0.67	0.034	1.8 · 10 ⁻¹⁰	0.47	0.024
Antenna thrust (100 W)	3.1 · 10 ⁻¹⁰	0.37	0.005	4.9 · 10 ⁻¹⁰	0.79	0.010	4.9 · 10 ⁻¹⁰	2.3	0.030
General relativity	2.8 · 10 ⁻¹⁰	0.33	0.004	2.1 · 10 ⁻¹⁰	0.33	0.004	7.1 · 10 ⁻¹¹	0.33	0.004
Venus (inf. conj.)	1.7 · 10 ⁻¹⁰	0.11	0.010	1.9 · 10 ⁻¹⁰	0.20	0.011	2.8 · 10 ⁻¹⁰	0.83	0.046
Ocean tides	1.2 · 10 ⁻¹⁰	0.10	0.009	7.5 · 10 ⁻¹¹	0.09	0.010	1.8 · 10 ⁻¹¹	0.13	0.009
Jupiter (opposit.)	2.3 · 10 ⁻¹¹	0.014	0.0014	2.5 · 10 ⁻¹¹	0.024	0.0018	3.6 · 10 ⁻¹¹	0.099	0.007
Pot. terms degree > 8	9.1 · 10 ⁻¹²	0.0054	0.0006	2.8 · 10 ⁻¹²	0.0022	0.0004	5.5 · 10 ⁻¹⁴	0.0009	0.0003
Mars (opposit.)	1.6 · 10 ⁻¹²	0.0011	0.0004	1.7 · 10 ⁻¹²	0.0016	0.0004	2.5 · 10 ⁻¹²	0.0067	0.0008



GPS DISPOSAL ORBITS

- Satellites no longer actively transmitting are transitioned to LADO (Launch and disposal operations) system
- When a GPS satellite is no longer useful it is boosted to a disposal/parking orbit 500 km above the constellation
- Over long term, resonance effects of sun, moon produce an increase in eccentricity which causes satellites to cross through operational constellations
- Recommendation was made to carefully circularize satellite orbits after boosting to disposal orbit but there are logistics issues with how much fuel is left and what to do with it.
- Simulations¹ predict non-operational GLONASS satellites will cross GPS operational altitude within 40 years.
- More recent study² identified possibility of finding more stable options to be used for Galileo

1. C.C. Chao, R.A. Gick, "Long-term evolution of navigation satellite orbits: GPS/GLONASS/GALILEO," *Advances in Space Research*, 34 (2004) 1221-1226

2. Alessi, E.M., Deleflie, F., Rosengren, A.J. *et al.* A numerical investigation on the eccentricity growth of GNSS disposal orbits. *Celest Mech Dyn Astr* **125**, 71–90 (2016).
<https://doi.org/10.1007/s10569-016-9673-4>

GNSS Constellation Comparison

	GPS ¹	Glonass ¹	Galileo ¹	BeiDou-3 (BDS-3) ²
Number of Satellites	24/32	24	24/30	24 MEO + 3 spares 3 GEO, 3 IGSO
Orbit Planes	6	3	3	
Satellites per plane	4	8	8/10	
Type of orbit	~circular	~circular (e<0.01)	Circular	
Inclination	55 deg	64.8 deg +/- 0.3 deg	56 deg	55 deg (MEO & IGSO) 0 deg (GEO)
Nominal radius	25,560 km	25,480 km	29,600.318 km	27,900 km 42,164 km (GEO, IGSO)
Orbital period	11 hr 58 min	11 hr 15 min 44 s +/- 5s	14 hr 4 min 42 s	~12 hr 53 min (MEO) ~24 hr (GEO & IGSO)
Ground track repeat	1 sidereal day	8 sidereal days	10 sidereal days	
Satellite spacing	Uneven	Walker 24/3/1	Walker 24/3/1+6 spares	



Walker Constellation

- Walker *T/P/F* - Constellation of T-satellites in circular orbits, P-orbital planes evenly spaced in right ascension, evenly spaced within the orbit planes with phasing offset between the planes = $360 \text{ deg } F/T$
- GLONASS: $a=25,480 \text{ km}$, **inclination=64.8 deg**, Walker 24/3/1
24 satellites, in 3 orbit planes (separated by 120 deg)
8 satellites within plane spaced at 45 deg
adjacent planes shift true anomaly by 15 deg
- Galileo: **$a=29,599.801 \text{ km}$** , inclination=56 deg, Walker 24/3/1 + 6 on-orbit spares
24 satellites, in 3 orbit planes (separated by 120 deg)
8 satellites within plane spaced at 45 deg
adjacent planes shift true anomaly by 15 deg
- BDS III MEO part: $a=27,900 \text{ km}$, inclination=55 deg, Walker 24/3/1 + GEOs, IGSO
24 satellites, in 3 orbit planes (separated by 120 deg)
8 satellites within plane spaced at 45 deg
adjacent planes shift true anomaly by 15 deg

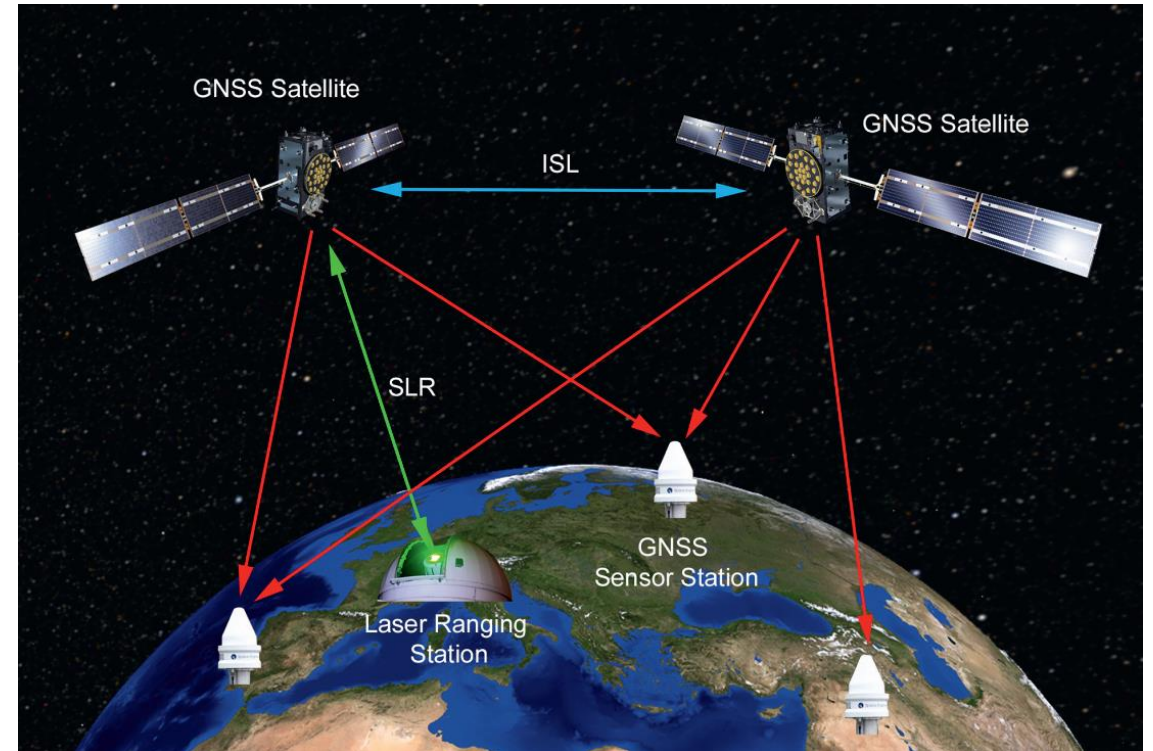


GPS EPHEMERIDES



GPS Satellite Orbit Determination

- Control segment collects range measurements at monitoring stations
- Uses precise orbit determination software including high fidelity models of perturbations to estimate position and velocity (and clocks and many other parameters) of the satellites
- **Representation of the orbit in the broadcast message** is a quasi-Keplerian parameterization that fits best over a 2-hour segment
 - toe – time of ephemeris,
 - IODE – issue of data ephemeris
 - Keplerian elements
 - 2 drift rates & Δn
 - 6 coefficients for harmonic corrections



[PNT book Figure 11.1] Orbit and clock determination of global navigation satellite systems is mainly based on one-way GNSS measurements (red) from a global station network connected to stable atomic clocks as a primary time reference. Complementary observation techniques may include inter-satellite links (blue) or ground-based satellite laser ranging (green).

GPS Satellite Orbit Representations/Ephemerides

- Three types of ephemerides used for computing GPS satellite positions with different accuracy and methods

YUMA Almanac

- Produced by GPS control segment
- Included in real-time navigation data
- Mean Keplerian elements
- Generated several times a week
- Accurate to several km
- Linear clock correction
- Used for quick visibility calc
- Available from navcen & celestrak.com

Broadcast Ephem

- Produced by GPS control segment
- Included in real-time navigation data
- Quasi-Keplerian elements
- New set every 2 hours
- Accurate to about 1 meter
- GPS satellite clock correction
- Used by GPS receivers to compute satellite positions
- Composite available from igs.org

Precise sp3

- Produced by IGS
- ECEF positions and satellite clock corrections
- Both rapid predictions and final products (diff latency)
- Accurate to 2.5 - 5 cm
- Used in precise applications
- Available from igs.org

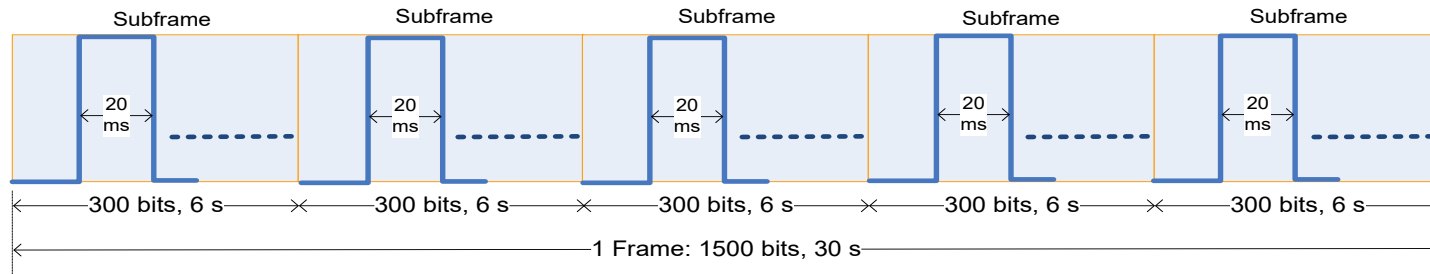
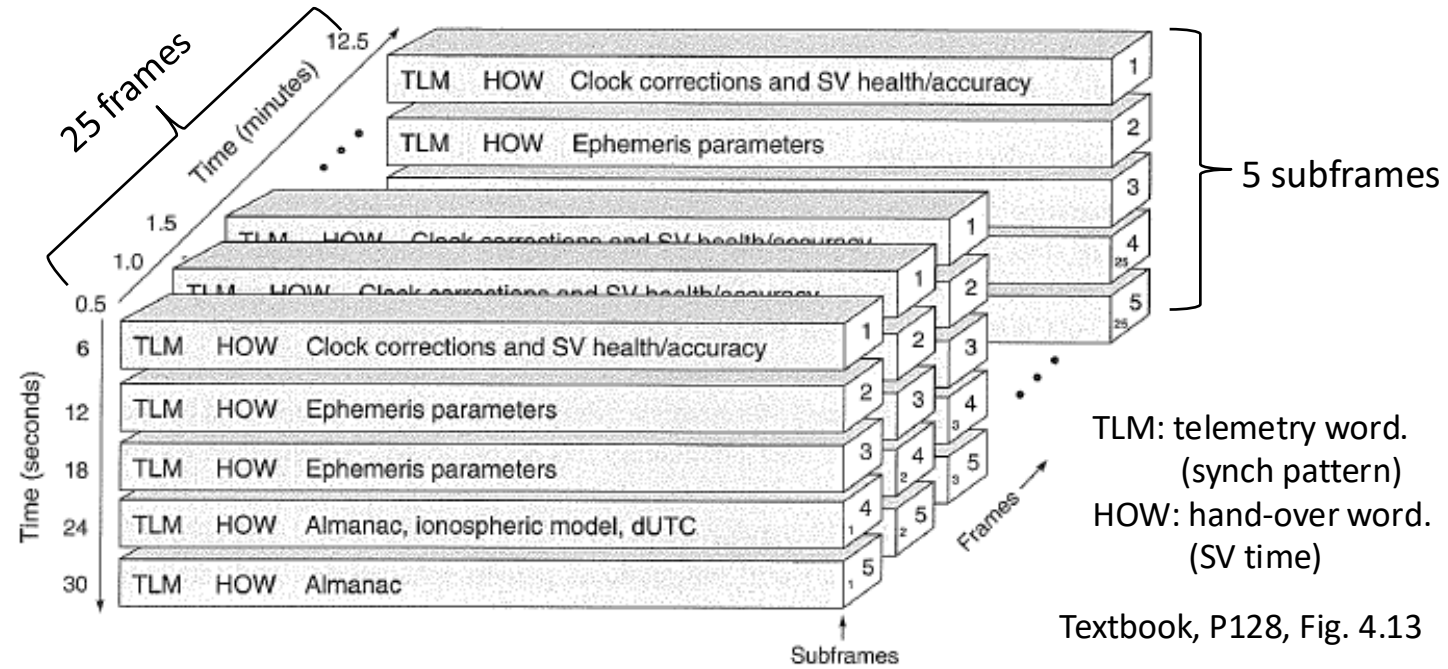
Example Almanac Data

```
***** Week 384 almanac for PRN-01 *****
ID:                                01
Health:                            000
Eccentricity:                      0.1892089844E-002
Time of Applicability(s):          503808.0000
Orbital Inclination(rad):          0.9569727161
Rate of Right Ascen(r/s):          -0.8034620388E-008
SQRT(A) (m 1/2):                   5153.544922
Right Ascen at Week(rad):          0.1254688426E+000
Argument of Perigee(rad):          0.168918399
Mean Anom(rad):                    0.2084193781E+001
Af0(s):                            0.1916885376E-003
Af1(s/s):                          -0.1091393642E-010

week:                              384
```



GPS Navigation Data Message



BROADCAST EPHEMERIS

M_0	Mean anomaly at reference time
Δn	Mean motion difference
$\sqrt{a}$	Square root of semi-major axis
Ω_0	Longitude of ascending node at weekly epoch
I_0	Inclination angle at reference time
ω	Argument of perigee
$\dot{\Omega}$	Rate of right ascension
e	Eccentricity
IDOT	Rate of inclination angle
C_{uc}	Cosine term, argument of latitude
C_{rc}	Cosine term, radius
C_{ic}	Cosine term, inclination
C_{us}	Sine term, argument of latitude
C_{rs}	Sine term, radius
C_{is}	Sine term, inclination
t_{oe}	Ephemeris reference time



$$\mu = 3.986005 \times 10^{14} \text{ m}^3 / \text{sec}^2$$

WGS84 (GPS) gravitational parameter for GPS ephemeris users

$$\dot{\Omega}_e = 7.2921151467 \times 10^{-5} \text{ rad} / \text{sec}$$

WGS84 (GPS) earth rotation rate

$$n_0 = \sqrt{\mu / a^3}$$

Computed mean motion

$$t_j = t - t_{oe}$$

Time from reference epoch (seconds)

$$n = n_0 + \Delta n$$

Corrected mean motion

$$M_j = M_0 + nt_j$$

Mean anomaly

$$M_j = E_j - e \sin E_j$$

Kepler's Equation

$$\tan \nu_j = \sqrt{1 - e^2} \sin E_j / (\cos E_j - e)$$

True anomaly

$$\phi_j = \nu_j + \omega$$

Argument of latitude

$$\delta u_j = C_{us} \sin 2\phi_j + C_{uc} \cos 2\phi_j$$

Argument of latitude correction

$$\delta r_j = C_{rs} \sin 2\phi_j + C_{rc} \cos 2\phi_j$$

Radius correction

$$\delta i_j = C_{is} \sin 2\phi_j + C_{ic} \cos 2\phi_j$$

Inclination correction

$$u_j = \phi_j + \delta u_j$$

Corrected argument of latitude

$$r_j = a(1 - e \cos E_j) + \delta r_j$$

Corrected radius

$$i_j = i_0 + \delta i_j + IDOTt_j$$

Corrected Inclination



ECEF GPS SATELLITE POSITION

Positions in orbital plane:

$$\begin{aligned}x'_j &= r_j \cos u_j \\ y'_j &= r_j \sin u_j\end{aligned}$$

Corrected longitude of ascending node:

$$\Omega_j = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e)t_j - \dot{\Omega}_e t_{oe}$$

Earth-fixed coordinates (ECEF):

$$\begin{aligned}x_j &= x'_j \cos \Omega_j - y'_j \cos i_j \sin \Omega_j \\ y_j &= x'_j \sin \Omega_j + y'_j \cos i_j \cos \Omega_j \\ z_j &= y'_j \sin i_j\end{aligned}$$



Precise Ephemeris – sp3 files

- NGA produces post-processed ephemerides based on GPS monitor & ref. stations measurements
- International GNSS Services (IGS) separately determines post-processed products based on independent global network – next Monday lecture will cover this

Products	Parameters	Accuracy	Latency	Update rates	Sample interval
Broadcast	Orbits	~160 cm	Real time		Daily
	Sat. clocks	~7 ns			
Ultra-rapid (predicted half)	Orbits	~10 cm	Real time	4 times/day	15 min
	Sat. clocks	~5 ns			
Ultra-rapid (observed half)	Orbits	<5 cm	3 hours	4 times/day	15 min
	Sat. clocks	~0.2 ns			
Rapid	Orbits	<5 cm	17 hours	Once/day	15 min
	Sat. clocks	0.1 ns			5 min
Final	Orbits	<5 cm	~13 days	Once/week	15 min
	Sat. clocks	<0.1 ns			5 min

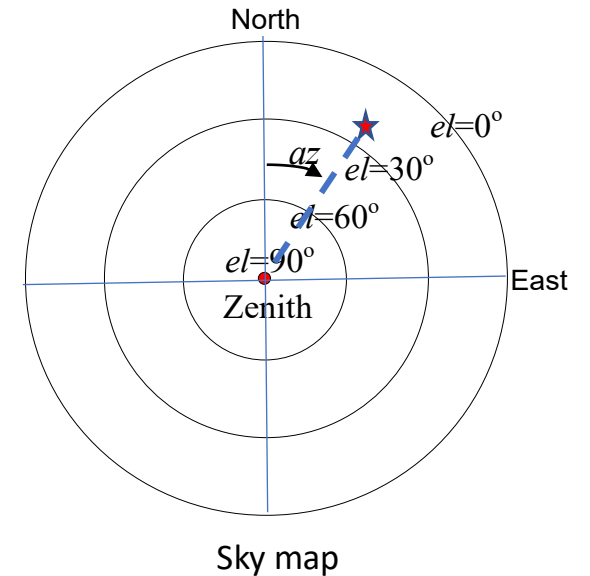
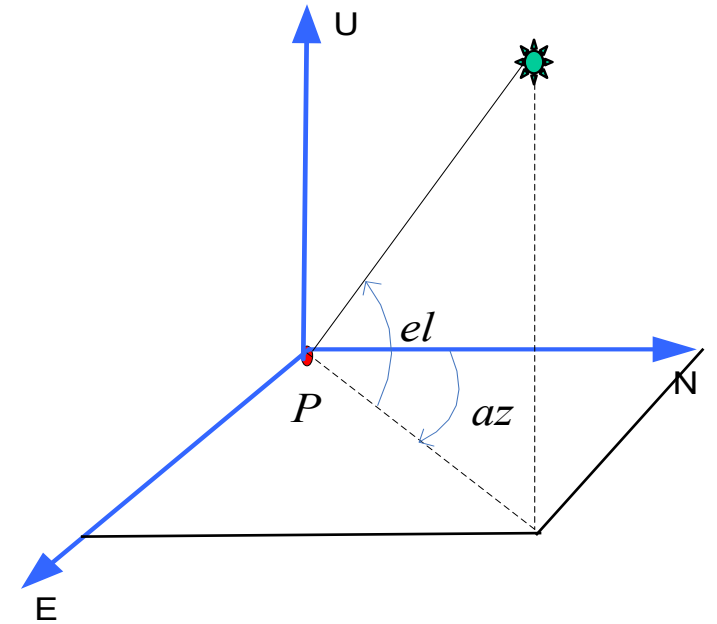


Visibility

- Assume you know observer location
- Compute position of satellite at time of interest
- Find unit vector pointing from observer to SV
- Rotate to the ENU frame

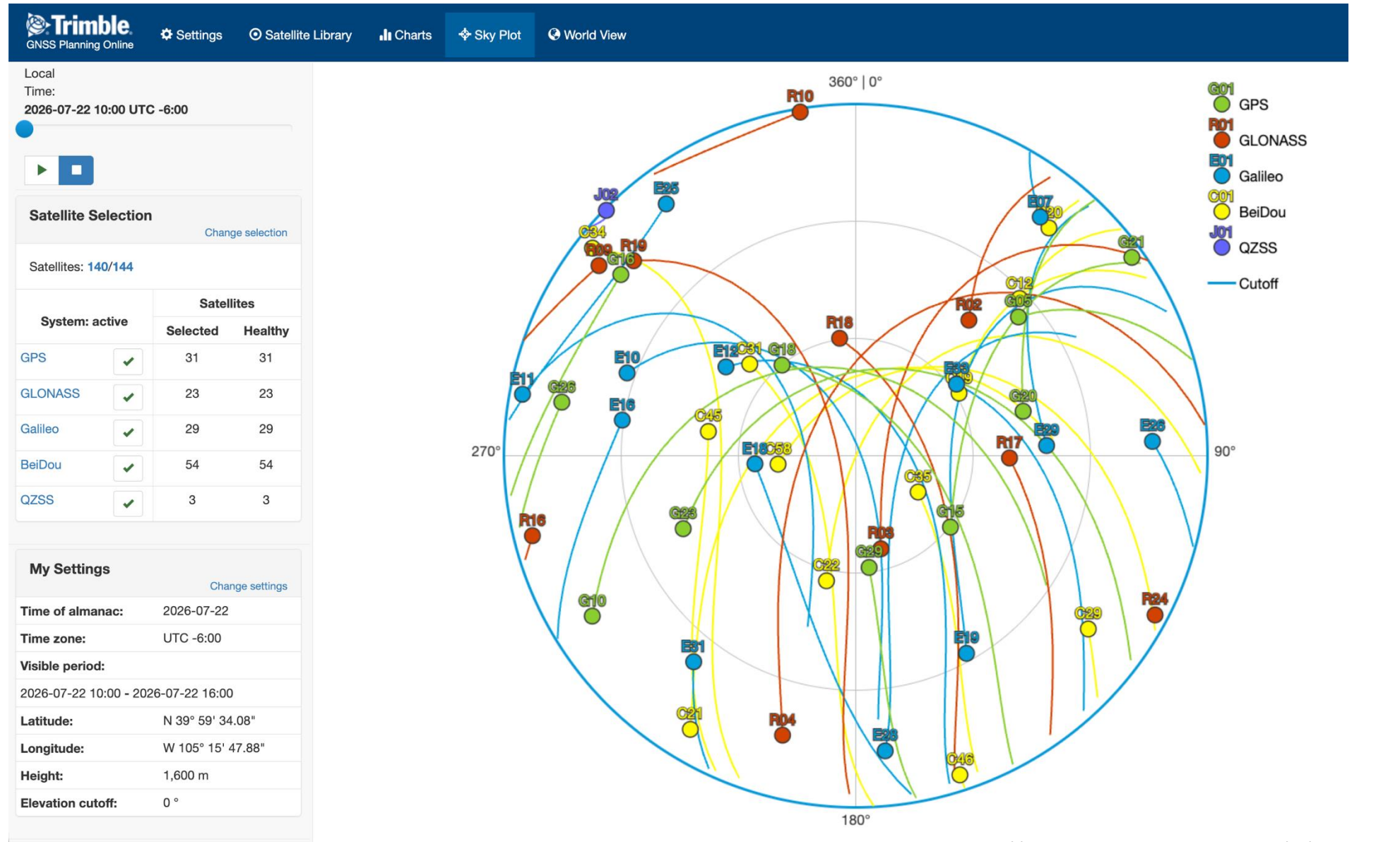
$$\tan az = \frac{x_E}{x_N} \qquad \sin el = \frac{x_U}{\sqrt{x_E^2 + x_N^2 + x_U^2}}$$

- Then check visibility based on elevation



GNSS Visibility

Boulder, CO
July 22, 2026
10:00 am MDT



GNSS ORBITS & VISIBILITY

Assigned: 4 September 2026

Due: 18 September 2026

OBJECTIVES

- Compute GPS satellite positions using an almanac and compare to precise positions from sp3
- Convert user positions between cartesian ECEF (x, y, z) and geodetic coordinates (lat, lon, height)
- Transform vectors between ECEF and local level coordinates
- Compute azimuth, elevation, and range from satellite and user positions
- Analyze satellite visibility

OVERVIEW

We will work with data for one day and several locations on the Earth:

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- User locations:
 - NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)
 - Smead Aerospace Backyard (choose reasonable coordinates)
 - EQUA (Latitude = 0, same longitude as NIST, choose a reasonable height)
- Almanac File: YUMA231.alm.txt (I added YUMA to the front of the filename)
Downloaded from <https://www.navcen.uscg.gov/archives> - on this website you choose the year & DOY
Or almanac.yuma.week0384.319488.txt
from <https://celestrak.org/GPS/almanac/Yuma/2026/> - on this website you choose the week number & TOA
- SP3 File: [IGS00PSFIN_20262310000_01D_15M_ORB.SP3](#)
Downloaded from CDDIS. *Caution – these are in km, all other positions in this HW are in meters.*
<https://cddis.nasa.gov/archive/gnss/products/2432/>

ASSIGNMENT

- 1) GPS Satellite Positions – **PRN05**
 - a. Read in the GPS satellite ECEF positions for the reference date from the sp3 file using the `read_sp3.m` function (or other means). Select the satellite specified above and plot its X, Y, Z components (in meters) vs time (GPS time shown as 0 to 24 hr into the day) as given in the sp3 file.
 - b. Now compute the GPS satellite ECEF positions for the same satellite at the same times, based on the YUMA almanac for that day. MATLAB functions for reading the file (`read_GPSSyuma.m`) and computing positions (`alm2pos.m`) are provided on Canvas – you may use these or other similar utilities, or write your own. Make sure to look through the code to see how the almanac algorithm is implemented. Add these calculated values to the plots produced in part a.
 - c. In a separate figure, plot the differences between the almanac and sp3 results (almanac minus sp3) vs time.
 - d. Briefly describe the accuracy of the GPS positions predicted by the almanac. Is this accuracy sufficient to determine if a satellite should be visible to an observer on the earth? Justify your answer.

2) Ground User Position and the ENU Frame

- a. Make a table of the ECEF positions and latitude, longitude, height (wrt WGS84 ellipsoid) for all three user locations. Please express all values in meters or degrees, with 1m precision. Explain your process, check your answers, and provide well-commented computer code.

Notes: `sprintf` and `fprintf` in MATLAB are useful for specifying the number of digits to print.

To do the conversion from ECEF to LLA, you can write your own function or use the MATLAB function `ecef2lla`. Check your result by converting your result back to ECEF. You can use `ecef2lla` to do that. Check the difference between the original and final ECEF positions.

- b. Write a function that takes as input a reference latitude and longitude in degrees and returns the transformation matrix to rotate from ECEF to ENU with respect to this reference location:

`C_ECEF2ENU = ECEF2ENU(ref_lat_deg, ref_lon_deg)`

Check your result for the NIST and EQUA locations given above : First find the reference position vector in ECEF and normalize to get its unit vector $[\hat{r}]_{ECEF}$. Then use the transformation matrix you found for this location to compute $[\hat{r}]_{ENU}$. Approximately what should it to be? Explain the results for both sites.

3) Single Satellite Visibility

- a. Write a function that computes the geometrical range (m), azimuth (deg), and elevation (deg) of a GPS satellite based on the ECEF locations of a user and GPS satellite. Check your function with a simple example or compare to a reliable source. (It would be helpful if this function could handle an array of GPS satellite positions.)
`[AZ, EL, RANGE] = compute_azelrange(userECEF, satECEF)`
- b. Calculate and plot the azimuth, elevation, and range vs time for the satellite chosen in part 1a as seen by a user (observer) at NIST. (Use separate panel for each one.)
- c. Repeat the plots in part 3b showing the values only when the satellite should be visible at NIST. Briefly describe the results.
- d. Compute and plot on the same graphs the results for a second satellite that has a different pattern of visibility than the original PRN. Briefly compare and explain why they might be different.

4) Full Satellite Visibility

- a. Calculate the azimuth and elevation angles and use `plotAzEl.m` or a similar utility to show the skyplot of all GPS satellites that should be visible on the reference day from NIST.
- b. Produce similar skyplots for the other two user locations specified above.
- c. Discuss the results and how the actual environment near the antenna would impact the visibility.

5) Optional - extra ideas for those interested in exploring further (not required, worth up to +5 extra points total)

- a. For ideal circular GPS orbits and a spherical Earth, derive analytically the maximum GPS satellite elevation an observer would see at the North Pole.
- b. For ideal circular GPS orbits and a spherical Earth, derive analytically the minimum GPS satellite elevation an observer at the Equator would see looking due North (i.e. at an azimuth of 0).
- c. Use the almanac to compute visibility for more than 1 day and determine the actual time necessary for the geometry to repeat.

Acknowledgements

ASEN5090 lecture materials have been developed at the University of Colorado Boulder by Professors Penina Axelrad, Dennis Akos, Kristine M. Larson, Jade Morton, and Yang Wang, with additional contributions by many outstanding TA's and students.

We thank Dr. Ben Bradley for significant contributions to this presentation.

HW 2 Posted

NEXT TIME – MEASUREMENT MODEL & IGS DATA

